

Bachelor Project

# AI Driven Analysis of Kidneys in CT Scans

## Weekly Report 7

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### What has been done this week

#### 1.1 Phase prediction

I have used the TotalSegmentator [1] tool for prediction of the contrast phase in CT scans, and obtained predictions on both the test set and the data set from Rigshospitalet.

##### 1.1.1 How the tool works

An explanation of the tool can be found in the TotalSegmentator GitHub repository: [https://github.com/wasserth/TotalSegmentator/blob/master/resources/contrast\\_phase\\_prediction.md](https://github.com/wasserth/TotalSegmentator/blob/master/resources/contrast_phase_prediction.md).

TotalSegmentator is used to predict masks for 22 different anatomical structures on the TotalSegmentator dataset. Most of these structures are found in the abdominal area. For each structure a median HU value is computed.

These values are then used as features to train an XGBoost (Extreme Gradient Boosting) regressor (**OBS: They write XGBoost classifier in the repository, but I guess they mean regressor as they are predicting continuous values?**), an implementation of gradient boosted decision trees. This model is actually an ensemble of five different models (**OBS: I think it is the regressor model that is actually an ensemble of five different regressors, but I feel like it is not completely clear from what they wrote in the repository.**

The regressor predicts the post injection time at imaging for a given CT scan, which can then be mapped to a phase by classification (**OBS: I don't think**

**they provided information on how this is done?)** into one of four classes: native, early arterial, late arterial, and portal venous.

It was found by 5-fold cross-validation that the F1 scores for the classes are 0.980 (native), 0.915 (early and late arterial), and 0.940 (portal venous).

Running the tool yields five outputs: predictions of the post injection time (pi.time) and the predicted phase, a probability, the standard deviation for the predictions derived from the five model ensemble, and corresponding bounds on the post injection time (pi.time\_min and pi.time\_max). The probability output is high if the predicted post injection time is close to the ideal post injection time for the predicted phase.

### 1.1.2 Results

In table 1 the predictions on the test set can be seen.

|       | pi.time | phase          | prob. | pi.time_min | pi.time_max | stddev |
|-------|---------|----------------|-------|-------------|-------------|--------|
| s0019 | 28.74   | arterial_early | 1.0   | 26.17       | 30.62       | 1.8680 |
| s0065 | 34.52   | arterial_late  | 1.0   | 29.40       | 37.39       | 3.2937 |
| s0371 | 38.82   | arterial_late  | 1.0   | 35.47       | 40.93       | 1.9624 |
| s0392 | 28.81   | arterial_early | 1.0   | 27.51       | 31.85       | 1.6275 |
| s0403 | 53.95   | arterial_late  | 0.7   | 51.67       | 57.26       | 1.8221 |
| s1285 | 26.19   | arterial_early | 1.0   | 24.08       | 27.46       | 1.1344 |
| s1314 | 66.15   | portal_venous  | 1.0   | 65.15       | 67.08       | 0.8138 |
| s1349 | 28.59   | arterial_early | 1.0   | 26.65       | 31.29       | 1.9458 |
| s1372 | 41.83   | arterial_late  | 1.0   | 38.47       | 46.89       | 2.7761 |
| s1382 | 44.00   | arterial_late  | 1.0   | 39.53       | 48.00       | 2.9300 |

Table 1: Phase predictions from TotalSegmentator on CT scans in test set.

When comparing to table 2, we see that the lower performance of the cortex-medulla segmentation framework on subject s1314 might be explained by the fact that the predicted phase for this subject is portal venous, where as all other subjects have prediction corresponding to an arterial phase.

|       | Background | Cortex | Medulla |
|-------|------------|--------|---------|
| s0019 | 0.9993     | 0.9044 | 0.8011  |
| s0065 | 0.9994     | 0.9114 | 0.8623  |
| s0371 | 0.9992     | 0.9036 | 0.8354  |
| s0392 | 0.9989     | 0.8723 | 0.8049  |
| s0403 | 0.9996     | 0.9163 | 0.7879  |
| s1285 | 0.9988     | 0.9009 | 0.8440  |
| s1314 | 0.9996     | 0.8479 | 0.5396  |
| s1349 | 0.9993     | 0.9138 | 0.7811  |
| s1372 | 0.9996     | 0.8502 | 0.8012  |
| s1382 | 0.9992     | 0.8886 | 0.8290  |

Table 2: DSC for background, cortex and medulla across subjects in test set.

The dataset from Rigshospitalet that I have been using contains 181 scans. Most of these were predicted to have late arterial phase. The exact number of phase predictions in each of the four classes can be seen in table 3

| native | arterial_early | arterial_late | portal_venous |
|--------|----------------|---------------|---------------|
| 0      | 1              | 164           | 16            |

Table 3: Number of phase predictions in each class for the RH dataset.

The probability outputs for the phases on the dataset were always either 1 or 0.7. For the early arterial phase and all portal venous phases the probability was 1. For the late arterial phases, 62 phases had probability 0.7, while 119 phases had probability 1.

## 1.2 Kidney cysts segmentation

I have run the kidney cysts subtask (which is the model with highest accuracy for segmentation of kidney cysts) from TotalSegmentator on the whole dataset from Rigshospitalet and saved the masks. I think the results look fine (but I am not an expert). An example can be seen in figure 1.

In total, cyst(s) on the left and/or right kidney were detected on 38 scans. Out of these 17 scans were of patients with low eGFR, 4 scans were of healthy patients, and the remaining 17 scans were of patients with high blood pressure.

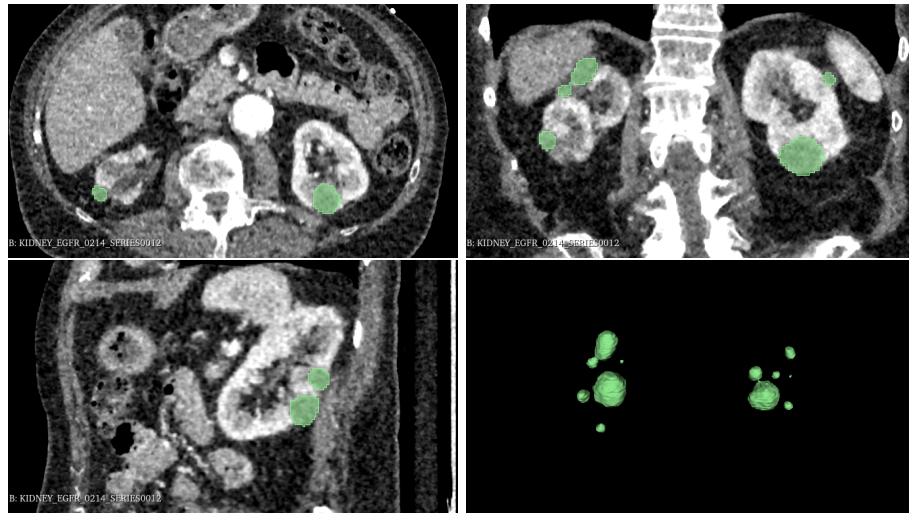


Figure 1: Prediction of cysts on scan of a patient with low eGFR from the RH dataset. The cysts are shown in axial, coronal, sagittal and 3D view (from left to right, top to bottom).

### 1.3 Cortex-medulla segmentation

Finally, I have run my cortex-medulla segmentation framework on the whole dataset from Rigshospitalet. By inspection of a few scan, the results looks promising. An example can be seen in figure 2.



Figure 2: Cortex (yellow) and medulla (blue) prediction on a scan from the RH dataset showing a patient with low eGFR. In the top row the predictions are seen in 3D view, while the predictions are seen in axial, coronal, and sagittal view (from left to right) in the bottom row.

However, the cortex-medulla segmentation sometimes overlaps with the kidney cysts segmentation. Therefore, I am going to modify my framework, such that voxels that are labelled as either cortex or medulla in the cortex-medulla segmentation mask, but is labelled as cyst in the kidney cysts segmentation mask, will have their labelled changed to background in the cortex-medulla segmentation mask.

For now, I have only modified the cortex-medulla mask in the described way for a single scan in the dataset. This is the scan, where predictions of cysts can be seen in figure 1. The cortex-medulla segmentation mask before and after label changing can be seen in figure 3.

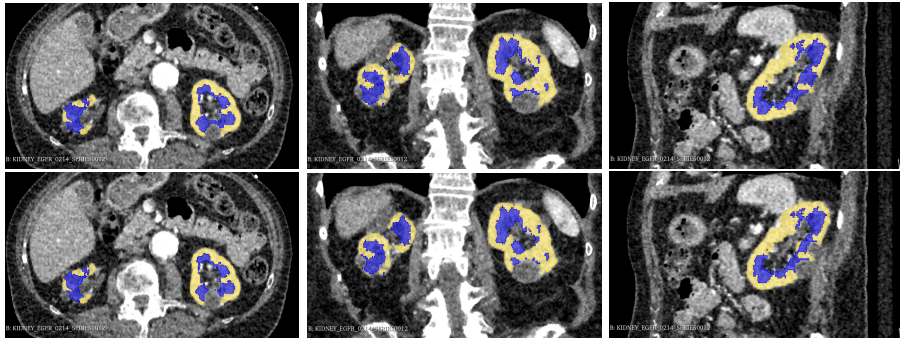


Figure 3: Cortex (yellow) and medulla (blue) prediction on a scan from the RH dataset showing a patient with low eGFR. In the top row the initial predictions are seen. In the bottom row the predictions after labelling voxels, that are predicted to be part of cysts, as background. Predictions are seen in axial, coronal, and sagittal view (from left to right).

## Project status according to the study plan

I think I am on track.

**Question:** I think the predictions of cortex and medulla look quite good, so I have not focused on trying to flip the images and see if that changes things. I don't know if it is worth the time to investigate that?

**Question:** I thought about running a kidney tumor segmentation model and change the cortex-medulla masks like I did using the cysts segmentations. However, I haven't found one published. I don't know if it is worth it to search more intensively for one and try to implement it, or if I should focus on extracting biomarkers from the cortex-medulla segmentations (modified in correspondence with detected cysts) I already have instead?

**Question:** I am a bit confused about exactly how the phase prediction tool works and whether what I wrote is correct (see my comments in section 1.1.1).

## Plan for the next weeks

- Modify the cortex-medulla framework and get segmentation masks without errors at cyst voxels for all scans in the RH dataset.
- Compute volume of kidneys, cortex, medulla, and volume ratios from predictions for data set from Rigshospitalet?
- Write sections for final report:
  - A complete technical background section (neural networks, convolutional neural networks, U-Net, maybe a bit about transformers and other methods I will use),
  - fit what I wrote about state-of-the-art into a complete section for the report,
  - a section describing the data I'm working with.
  - a section describing the implemented frameworks/methods/experiments
- Think about the structure of the final report (methods, model/framework, experiments etc.)

## References

- [1] Jakob Wasserthal, Hannah-Charlotte Breit, Michael T. Meyer, Matthias Pradella, Daniel Hinck, Andreas W. Sauter, Tobias Heye, Daniel T. Boll, Jobin Cyriac, Shuqing Yang, Matthias Bach, and Maximilian Segeroth. TotalSegmentator: Robust segmentation of 104 anatomic structures in CT images. *Radiology: Artificial Intelligence*, 5(5):e230024, 2023.